21-242 Recitation 12

Fields and Vector Spaces — Q2. $0 =$

Course-level solution, with a later-course alternative when useful

October 1, 2026

Prompt

Prove that $0\vec{v} = \vec{0}$.

Course-level solution

Let $\vec{v} \in V$. Then

$$0\vec{v} = (0+0)\vec{v} = 0\vec{v} + 0\vec{v}$$

by distributivity. Add the additive inverse $-(0\vec{v})$ to both sides:

$$\vec{0} = 0\vec{v}.$$

(Equivalently: $0\vec{v} + 0\vec{v} = 0\vec{v}$ and Q4 below, or cancellation.)

Later-course solution (using ideas from the rest of the handout)

In coordinates relative to any basis $\mathcal{B}$, we have $[0\vec{v}]_{\mathcal{B}} = 0[\vec{v}]_{\mathcal{B}} = \vec{0}$ in $\mathbb{F}^n$, hence $0\vec{v} = \vec{0}$. This uses the coordinate isomorphism, which is later in the handout.